Periodic Center-Flux Carriers on the 2^4 Cubical Torus

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Abstract

We replace the topology-free open block by an exact periodic $GF(2)$ cochain carrier and construct the six independent center two-flux labels required by a sector-aware retained state.

Let C^k be periodic cubical cochains on $(\mathbb{Z}/2\mathbb{Z})^4$ with $GF(2)$ coefficients. The executable matrices include all coboundaries d_0, d_1, d_2, d_3.

Theorem 1. *The cell counts are* $(16, 64, 96, 64, 16)$*, the coboundary ranks are* $(15, 45, 45, 15)$*, and*

$$d_{k+1}d_k = 0, \qquad (b_0, b_1, b_2, b_3, b_4) = (1, 4, 6, 4, 1).$$

Six explicit cut-sheet two-cocycles are independent modulo $\operatorname{im} d_1$ *and therefore label* $2^6 = 64$ *center-flux sectors.*

Proof. Gaussian elimination over $GF(2)$ gives the ranks. Rank-nullity gives the Betti numbers. Appending the six representatives to d_1 raises its rank from 45 to 51, while multiplication by d_2 vanishes. □

These labels can be retained through a periodic block exposure and prevent silent identification of distinct center sectors.

Remark 1. *The construction represents* $GF(2)$ *center two-flux, not the integer second Chern class or instanton number of a smooth* $SU(2)$ *bundle. It is a topology carrier, not complete topological control.*